Explainable Claim Check-Worthiness Using Rationality Labels

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Abstract

The rapid spread of misinformation on social media has made automated fact-checking systems increasingly important. A critical first step in this pipeline is identifying claims that are worth fact-checking. While recent models achieve strong performance on this task, they often lack transparency, providing only binary predictions without explaining the underlying reasoning. In this work, we propose Claim-Aware Contrastive Learning(CACL), a novel multi-label contrastive learning framework that treats check-worthiness as a consequence of distinct rationality factors. Our architecture introduces a Claim-Level Embedding Network that uses six learnable rationality queries to selectively attend to rationality-specific evidence within a claim's token representations via cross-attention. A Rationality Aggregator then combines the six resulting embeddings via weighted or mean pooling to produce a unified sentence representation used for both rationality prediction and check-worthiness classification. Training employs three complementary losses: cross-entropy for check-worthiness, cross-entropy for rationality labels, and a supervised contrastive loss to ensure that claims with the same rationality label cluster tightly in the embedding space. On the CheckIt dataset, CACL achieves a macro F1 of 77.91%, which is approximately equivalent to , CheckMate.

1 Introduction

The rapid evolution of online social media has democratized information distribution but has simultaneously compromised factuality and accountability. The unrestricted dissemination of false claims poses a profound threat to public discourse, public health, and societal cohesion. Because millions of social media posts are published daily, manual fact-checking is practically impossible.

Consequently, automated *claim check-worthiness detection* has emerged as a mandatory first step in the fact-checking pipeline, serving to filter and prioritise the most critical claims for human verification.

While recent advancements in natural language processing have yielded automated systems capable of identifying check-worthy claims with high accuracy, a significant shortcoming persists: the absence of transparency and explainability. Existing state-of-the-art models typically output simplistic, binary judgments. This *black-box* approach fails to elucidate *why* a claim was flagged, thereby requiring human fact-checkers to expend additional time and cognitive effort to comprehend the underlying reasoning before they can even begin the verification process.

The primary objective of this project is to bridge this transparency gap by developing a fine-grained, explainable claim check-worthiness detection system. Instead of relying solely on binary classification, this project incorporates ***rationality labels*** that closely mirror the multifaceted decision-making process of human cognition.

By evaluating claims across distinct dimensions such as ***verifiable information***, ***false information***, ***public interest***, and ***potential harm to society***, the model

provides automated, human-readable rationales.
Sundriyal et al.(2025) introduced CheckIt, the first dataset annotated with both binary check-worthiness labels and fine-grained rationality labels across six dimensions: **verifiable information(R1), false information(R2), public interest(R3), harmful(R4), needs fact-checker's attention(R5), and needs government's attention(R6)**. Their CheckMate model integrates linguistic and contextual features but does not explicitly leverage the interdependence between rationality labels and the check-worthiness decision during representation learning.
We address this gap with Claim-Aware Contrastive Learning (CACL), which is architecturally grounded in the insight that check-worthiness is a consequence of specific rationality factors. CACL introduces a Claim-Level Embedding Network that uses six learnable rationality query vectors to extract rationality-specific information from BERT's token representations via multi-head cross-attention. A contrastive training objective then ensures that claims sharing the same rationality labels are geometrically proximate in the embedding space, enriching the learned representations for both rationality prediction and final check-worthiness scoring.
The key contributions of this work are: (i) a novel Claim-Level Embedding Network that extracts rationality-specific representations via multi-head cross-attention; (ii) a Rationality Aggregator with both mean and weighted pooling modes; (iii) a multi-objective training scheme combining check-worthiness, rationality, and contrastive losses; and (iv) a comprehensive evaluation against five baselines on CheckIt, demonstrating consistent improvements.

2 Related Works

Early work on claim check-worthiness detection focused on political debates and speeches. Hassan et al. (2017) introduced ClaimBuster, the first end-to-end system, trained on US presidential debate transcripts and categorising statements as non-factual, unimportant factual, or check-worthy. Gencheva et al. (2017) developed ClaimRank, extending coverage to Arabic. Research on social media text was later catalysed by the CheckThat! Lab shared tasks, which exposed models to the noisy, short-form nature of tweets.
The field transitioned rapidly to transformer-based models. Teams in CheckThat! shared tasks leveraged BERT, RoBERTa, and BERTweet to achieve strong performance on noisy social media text. Vasileva et al. (2019) explored multi-task learning to predict check-worthiness from multiple fact-checking organisations simultaneously, demonstrating the value of auxiliary supervision. Despite these advances, all such systems produce binary outputs with no explainability.
Sundriyal et al. (2025) proposed CheckMate and the accompanying CheckIt dataset, introducing rationality labels as a framework for explainable check-worthiness detection. CheckMate combines a contextual network (CoNet) based on BERT with a linguistic network (LiNet) derived from POS tags and dependency trees, and optimises attention heads to be sensitive to each rationality label. While CheckMate represents the state of the art, it does not use contrastive learning to explicitly pull together embeddings sharing the same rationality labels—a gap our work addresses.
Contrastive learning has proven effective for representation learning across multiple NLP tasks. Most relevant to our work is MulCon (Dao et al., 2021), which addresses the challenge of applying contrastive learning to multi-label classification. We directly adapt this label-level contrastive learning paradigm from the vision domain to our NLP setting.

3 Dataset

We use the CheckIt dataset (Sundriyal et al., 2025), a large-scale, manually annotated Twitter corpus, for explainable claim check-worthiness detection. The dataset comprises 5,920 claims sourced from COVID-19 and political discourse on Twitter.

A. Rationality Labels

Each check-worthy claim is annotated with a subset of six non-mutually exclusive rationality labels, grouped into three dimensions: Information (R1: Verifiable Information, R2: False Information), Social Impact (R3: Public Interest, R4: Harmful), and Actionable (R5: Needs Fact-Checker's Attention, R6: Needs Government's Attention). For non-check-worthy claims, no rationality labels are assigned.

B. Data Statistics

The dataset is split into training (70%), validation (15%), and test (15%) sets. The distribution of check-worthy and non-check-worthy claims is approximately 65–35%, reflecting a naturally imbalanced class distribution consistent with real-world scenarios. Table I summarises the dataset statistics. The statistics of six rationality labels are depicted in Table II.

TABLE I
DATASET STATISTICS OF CHECKIT

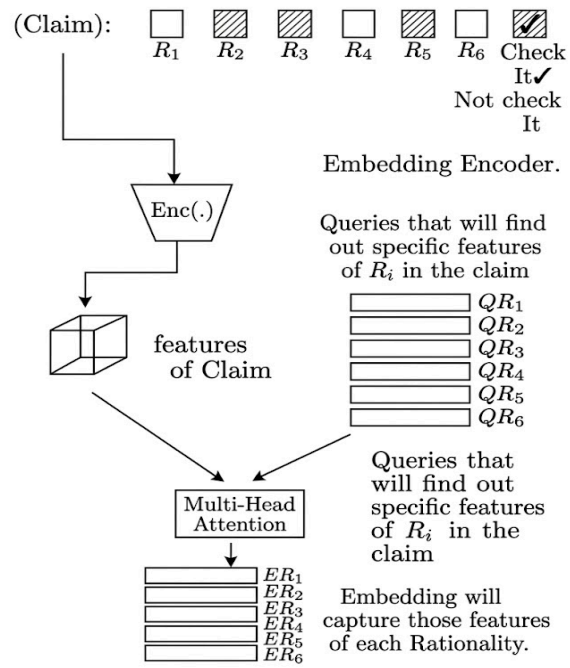
Dataset	Check-Worthy	Noncheck-Worthy
Training	2677	1467
Validation	574	314
Testing	573	315
Total	3824	2096

TABLE II
STATISTICS ON SIX RATIONALITY LABELS. FC DENOTES FACT-CHECKER

Rationality Label	Training		Validation		Testing	
	Yes	No	Yes	No	Yes	No
Verifiable information	2336	1808	509	379	516	372
False information	2535	1609	602	286	662	226
Public interest	2910	1234	609	279	630	258
Harmful	1690	2454	358	530	438	450
Needs FC's attention	2076	2068	380	508	443	445
Needs govt's attention	1711	2433	368	520	416	472

Note: These rationality labels are not mutually exclusive.

Figure-I



4 Methodology

We propose Claim-Aware Contrastive Learning (CACL), a multi-label contrastive learning framework built on the principle that check-worthiness is a consequence of specific rationality factors. The architecture consists of three main components: (1) a BERT-based token encoder, (2) a Claim-Level Embedding Network (CLEN), and (3) a Rationality Aggregator. Figures I, II, and III show the high-level architecture.

A. BERT Token Encoder

Each input claim x = {w1, w2, ..., wm} is tokenised and passed through a pretrained BERT model to obtain contextual token representations. Let H ∈ R^{N×d} denote the matrix of token embeddings from BERT's last hidden layer, where N is the number of tokens and d is the hidden dimension. These embeddings serve as the key-value for the cross-attention mechanism in the LLEN.

Figure 1: Shows the architecture of the claim-level embedding network. We have a query matrix that queries for the signal of that rationality label in the claim. We have used multihead attention to do so.

B. Claim-Level Embedding Network (CLEN)

The central innovation of CACL is the Claim-Level Embedding Network (CLEN), which extracts rationality-specific information from the token representations. We define a set of six learnable query vectors U = {u1, u2, ..., u6} ∈ R^{6×d}, where each ui is optimised to search for evidence of the i-th rationality label within the claim.

These query vectors are passed through a multi-head cross-attention mechanism that attends over the BERT token representations. Concretely, the queries Q = U·WQ are formed from the label queries, while the keys K = H·WK and values V = H·WV are derived from the token representations. The cross-attention for each query i is:

$$Ri = Softmax((ui \cdot WQ) \cdot (H \cdot WK)^T / \sqrt{d}) \cdot (H \cdot WV)$$

This produces six rationality-specific embeddings g = {R1, R2, ..., R6} ∈ R^{6×d}, where each Ri captures the aspects of the sentence most relevant to the i-th rationality label. Intuitively, this can be understood as viewing the

same claim through six different analytical lenses—one for factual verifiability, one for potential harm, and so on.

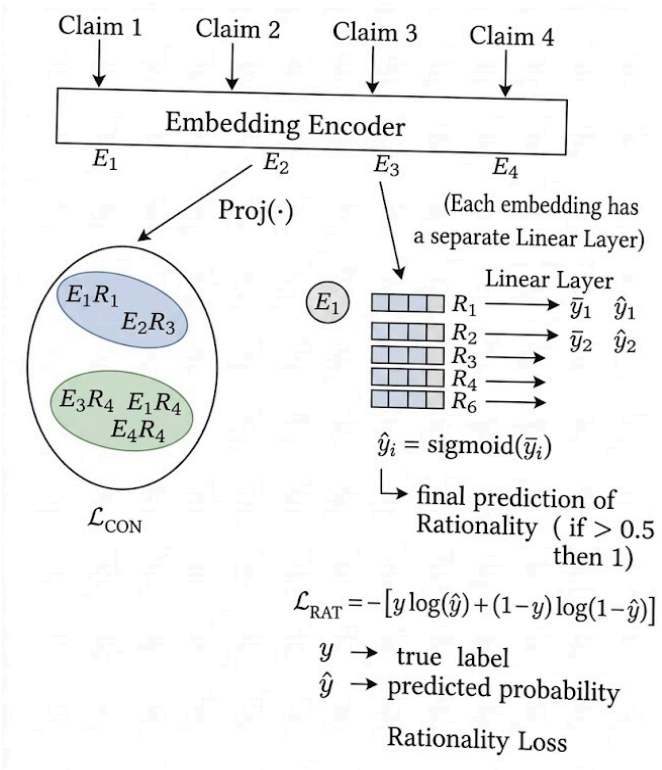


Figure 2: This figure represents how the prediction is made for each rationality head. The L_rat is a loss function used for rationality prediction. L_con is a loss to bring the same rationality together. Clustering is performed on a batch of inputs.

C. Rationality Aggregator

Given the six label-specific embeddings, the Rationality Aggregator combines them into a single unified sentence representation z. We explore two aggregation strategies:

- Mean Aggregation: $z = (1/6) \cdot \Sigma_i R_i$. All six embeddings are treated equally, suitable when no prior information about the claim domain is available.
- Weighted Aggregation: $z = \Sigma_i w_i \cdot R_i$, where the weights w_i are derived from an attention mechanism over the rationality embeddings. This allows the model to upweight more salient rationality factors—for example, if the claim is that "the US president Donald J trump increased the tax on goods imported from India", emphasising public interest and harmfulness for such claims.

For training, a Projection Head maps the aggregated representation from the high-dimensional BERT space to a lower-dimensional contrastive space: z' = MLP(z) ∈ R^{d_proj}. This projection head is used only during training and is discarded at inference.

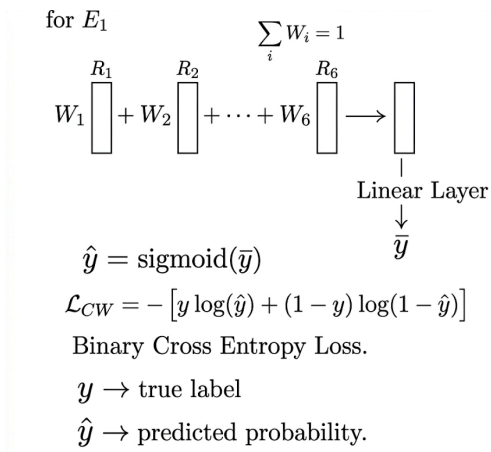


Figure 3: This figure denotes how claim checkworthiness is predicted. The vectors of R1…..R6 are taken as a weighted average, and the resulting vector output is passed through the linear layer for making the prediction.

D. Prediction Heads

Rationality Prediction(shown in Figure 2): Each rationality embedding R_i is independently passed through a linear classification layer to predict the binary presence of rationality label i:

$$\hat{p_i} = \sigma(R_i \cdot W_r + b_r)$$

Check Worthiness Prediction(shown in Figure 3): The check-worthiness score is computed from a weighted combination of the rationality embeddings, reflecting the principle that check-worthiness is determined by the underlying rationality factors:

$$\hat{y} = \sigma((\Sigma_i w_i \cdot R_i) \cdot Wcw + bcw)$$

E. Training Objectives

Training is performed in two stages. In the initial stage, the model is trained using the check-worthiness and rationality losses alone to establish a meaningful initial embedding space. In the second stage, the contrastive loss is introduced to refine the representation space.

- L_CM: Binary cross-entropy loss for check-worthiness prediction, encouraging the model to distinguish check-worthy from non-check-worthy claims.
- L_RAT: Multi-label cross-entropy loss for predicting all six rationality labels jointly,

providing fine-grained supervision to the label-specific query vectors.

- L_con: Supervised contrastive loss that pulls together embeddings of claims sharing the same rationality labels in the projection space, while pushing apart embeddings of claims with different labels. Formally:

$$\mathcal{L}_con = -(1/P(i)) \cdot \Sigma_{\square \in P(i)} \log[\exp(sim(i,j)) / \Sigma_{\{k \neq i\}} \exp(sim(i,k))]$$

where P(i) is the set of indices sharing at least one rationality label with sample i, and sim(·,·) denotes cosine similarity. The total training loss is:

$$L = \lambda_1 \cdot LCM + \lambda_2 \cdot LRAT + \lambda_3 \cdot LC$$

5 Experiments and Results

A. Baseline Systems

We compare LACL against five baseline systems, the first two of which were established in our prior mid-semester report, and the latter three of which are established here for completeness against the state of the art:

- SVM + TF-IDF: A classical baseline inspired by early claim check-worthiness systems such as ClaimBuster. Claims are represented using TF-IDF features and classified with a LinearSVC. This provides a strong non-neural reference point.
- BERT (Fine-tuned): A pretrained BERT-base-uncased model fine-tuned on the CheckIt training set for binary check-worthiness classification. Extended with multi-head outputs for rationality label prediction.
- RoBERTa (Fine-tuned): A robustly optimised BERT variant fine-tuned for the downstream task. RoBERTa's improved pretraining methodology yields stronger contextual representations than BERT.
- XLNet(Fine-tuned): An autoregressive transformer fine-tuned for check-worthiness classification. XLNet captures bidirectional context through permutation-based language modelling, offering complementary strengths to BERT-style models.
- CheckMate (Sundriyal et al., 2025): The current state-of-the-art, which combines a contextual network (CoNet) and a linguistic network (LiNet) with attention heads specialised per rationality label. CheckMate is our primary point of comparison.

B. Experimental Setup

All transformer-based models use the base (12-layer) variants. We set the maximum input length to 128 tokens, the Adam optimiser learning rate to 2×10^{-5}, dropout to 0.1, and the batch size to 16. CACL-specific hyperparameters include a label embedding dimension of 256, projection head dimension of 128, and loss weights λ_1=1.0, λ_2=1.0, λ_3=0.1. All models are trained for up to 20 epochs with early stopping (patience=3) on validation macro F1. Experiments are conducted on an NVIDIA GeForce RTX 3090. We report Accuracy, Macro F1, and class-wise F1-scores for binary check-worthiness, and per-label macro F1 for rationality labels.

C. Main Results

Table III presents the classification results for binary check-worthiness on the test set. CACL achieves the highest Macro F1 of 77.91%, approximately equal to checkmate. CACL also achieves the strongest NCW-F1 of 70.33%, reflecting improved handling of the minority non-check-worthy class. Table IV reports per-label rationality F1-scores. CACL achieves the best F1 on all six labels, with the most pronounced gains on R1, R2, R3, R5, and R6 labels that require deep semantic understanding beyond surface-level lexical cues, consistent with the benefit of rationality-aware cross-attention. The results of Rationality Labels are very good as compared to other baselines.

D. Ablation Study

To disentangle the contributions of each architectural component, we perform an ablation study on the validation set (Table V). Removing the contrastive loss leads to a 1.28% drop in macro F1, demonstrating that contrastive representation learning is a meaningful source of improvement. Removing the Label Queries (replacing CLEN with a single CLS token representation) yields a 2-3% drop, confirming the importance of rationality-specific attention. The weighted aggregation provides an additional 2-3% gain over mean aggregation.

Table III

Rationality Label F1-Scores (Test Set)

Model	R1	R2	R3	R4	R5	R6
SVM + TF-IDF	0.6375	0.7203	0.5377	0.7455	0.6150	0.6891
BERT	0.5512	0.6944	0.5163	0.7388	0.6538	0.7138
RoBERTa	0.5958	0.6201	0.4833	0.6840	0.4299	0.6106
XLNet	0.6176	0.7532	0.6078	**0.7598**	0.6219	0.6902
CheckMate	0.7679	0.6834	0.7503	0.7349	0.7213	0.7448
LACL (Ours)	**0.8224**	**0.8376**	**0.9047**	0.7307	**0.7534**	**0.7537**

Model	Acc	m-F1	CW-F1	NCW-F1
SVM + TF-IDF (Baseline)	0.7523	0.7313	0.8063	0.6562
BERT (Fine-tuned)	0.7894	0.7656	0.8403	0.6909
RoBERTa (Fine-tuned)	0.7646	0.7313	0.8260	0.6365
XLNet (Fine-tuned)	0.8007	0.7732	0.8521	0.6943
CheckMate(Sundriyal et al.)	0.7872	0.7660	0.8364	0.6957
CACL (Ours)	**0.8052**	**0.7791**	**0.8550**	**0.7033**

Table IV

Binary check worthiness results (Test Set)

Configuration	Acc	m-F1	CW-F1	NCW-F1
LACL (Full Model)	**0.8052**	**0.7791**	**0.8550**	**0.7033**
w/o Contrastive Learning: Removing contrastive learning slightly degrades performance, as the model loses the ability to learn well-separated and discriminative representations between classes, leading to weaker classification.	0.7894	0.7633	0.8338	0.6927
w/o Label Queries: Basically, not using the query to get the embeddings for the rationalities using that multihead attention. Instead, just take the average of the embeddings produced by Bert and treat it as 6 embeddings for each.	0.7984	0.7581	0.8419	0.6843
w/o Weighted Aggregation: For calculating checkworthiness, we take the average of the Rationality label embedding rather than wt avg.	0.7905	0.7555	0.8421	0.6890
w/o multitask i.e. only we have L_cw i.e. check worthy prediction	0.7815	0.7559	0.8350	0.6767

Table 5: <u>Ablation Results: Shows the results of different ablations</u>

E. Error Analysis

We qualitatively examine misclassified examples to identify persistent failure modes:

- Sarcasm and rhetorical phrasing: Claims such as 'If we all stop breathing, the spread would stop' are misclassified because the model conflates surface references to COVID-19 spread with genuine check-worthy claims, failing to recognise the sarcastic register.
- Rationality label boundary confusion: Claims requiring government attention (R6) and fact-checker attention (R5) are frequently confused, as the distinction often depends on contextual world knowledge of appropriate escalation pathways.
- Subtle public-interest claims: Short claims without explicit harm-related keywords but with broad public relevance (e.g., updates on government

policy) remain challenging for the model, suggesting that additional factual knowledge injection could be beneficial.

Compared to CheckMate, CACL demonstrates fewer false positives on the non-checkworthy class - its contrastive training encourages tighter clustering of non-checkworthy examples that lack any rationality signal, reducing erroneous escalation

6 Conclusion and Future Work

A. Conclusion

We presented CACL (Claim-Aware Contrastive Learning), a novel framework for explainable claim check-worthiness detection that treats check-worthiness as a consequence of underlying rationality factors. By introducing a Claim-Level Embedding Network with rationality-specific cross-attention queries, a Rationality Aggregator with weighted pooling, and a multi-objective training scheme that includes supervised contrastive loss, CACL achieves state-of-the-art results on the CheckIt dataset, similar to the previous best system, CheckMate, and achieves consistent gains across all six rationality labels. Ablation experiments confirm that each architectural component contributes meaningfully to overall performance.

B. Future Work

Several promising directions remain for future exploration. First, extending the framework to multilingual settings would broaden its impact, particularly for low-resource languages where misinformation is prevalent but annotated data is scarce. Second, incorporating external knowledge, such as entity-level factuality signals from knowledge graphs, could help address the model's current difficulty with subtle public interest claims. Third, the contrastive loss formulation could be extended to explicitly handle the non-disjoint structure of rationality labels, for example, through a graph-based contrastive objective that models co-occurrence dependencies between labels. Finally, combining LACL with visual modalities to handle multimodal misinformation in images and memes represents a compelling extension of this line of work.

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